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A radiation shielding sensitivity analysis based on metal hydrides multilayers

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Abstract

In order to shield neutron and gamma rays efficiently, a multilayer model is designed with metal hydrides and heavy metals and is analysed based on Monte Carlo simulations. In terms of shielding performance, the hydrogen in metal hydrides acts as a moderator to slow down the neutron energy and heavy metals are good for absorbing gamma rays. A simulation and calculational analysis are carried out with various parameters such as spectrum change, shield thickness, and number of multilayers. In addition, the rate of DPA (displacement per atom) is analysed to estimate both the lifetime and radiation resistance with the MCNP code. From lots of simulations, ZrH_2 and W couples are the best candidate especially for shielding gamma rays, while TiH_2 with W is good for neutron shielding. The concept of multilayer metal hydride such as TiH_2 and ZrH_2 coupled with W could be one of best combinations to shield both neutron and gamma-rays in many nuclear facilities such as nuclear reactor, fusion reactor and other applications.

Keywords: metal hydrides, shielding materials, neutron, gamma, multilayer, Monte Carlo, DPA

(Some figures may appear in colour only in the online journal)

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1. Introduction

From the beginning of nuclear era, radiation shielding has become an essential part for the design of a nuclear reactor, a particle accelerator, spacecraft, and other various nuclear facilities (Fantidis 2015). Various shielding materials have been proposed and most shielding materials have highly effective radiation resistant properties including challenging under high temperature, a reduced corrosion in extreme environments, and cost effectiveness. Therefore, a profound selection of these materials is important to ensure a long lifetime in the various applications (Hu *et al* 2008). Hu *et al* (2008) and Fantidis (2015) have reported that the one of effective shielding materials for nuclear reactors is mixed hydrogenous materials, heavy metal elements, and other neutron absorber materials. The elastic scattering of hydrogen and inelastic scattering of heavy metal elements work effectively decelerating fast neutrons to slow down. And the absorbers contribute to reduce significantly secondary gamma rays as well as thermal neutrons (Hu *et al* 2008, Hayashi *et al* 2009, Cossairt 2014).

Table 1 lists the metal hydrides and single high-Z atomic number metals as the candidate shielding materials for various previous researches. The material densities and weight fractions are presenting as well. The possible advanced shielding materials include lithium hydride (LiH), magnesium hydride (MgH_2), titanium hydride (TiH_2), and zirconium hydride (ZrH_2), lead (Pb), Tantalum (Ta) and Tungsten (W). Especially, LiH is an attractive material for serval applications including the fuel cell, hydrogen storage or neutron shielding in nuclear industry (Montgomery 1973; Welch 1974, Goertz *et al* 1995, Wang *et al* 2013, 2016a, 2016b; Weber *et al* 2018). For neutron shielding in space reactors, LiH has been tested with a low density of 0.775 g cc^{-1} at room temperature and reveals as an efficient neutron shielding material (Welch 1974). Therefore, LiH will be used as one of the shielding material in this paper. Another material such as MgH_2, is outlined to be one of the best candidates for the development of light metal hydride-based hydrogen storage materials (Eigen, 2007). The inexpensive and available metal such as Magnesium can produce MgH_2 by the reaction of $Mg(OH)_2$ with water (Lukashev *et al* 2008). Titanium hydride (TiH_2) has a relatively high neutron absorption cross section, therefore it is an interesting shielding material to reinforcing other materials (William *et al* 1968). TiH_2 has also reported to show superior radiation shielding capability than stainless steel SS316 L mixed water, which has led to a reduction in the thickness of neutron shielding in fusion reactors (Hayashi *et al* 2006, 2009). In addition, ZrH_2 has been demonstrated to be similar shielding capabilities as TiH_2 (Hayashi *et al* 2006, 2009). High atomic number metals, high Z metals, such as Tungsten (W) and Tantalum (Ta) are good candidate for gamma ray shielding because it is known that the reaction of gamma rays is proportional to the Z (Lamarsh, *et al* 2001). For example, photoelectric cross sections depend on strongly on Z^n and the pair production cross section is a function of Z such as Z^2. And Compton scattering is also dependent on Z. The dense metals such as W and Ta have high melting points, 3410 °C for W, and 3017 °C for Ta and exhibit endurance of mechanical and thermal stresses. However, W and Ta are slightly expensive, and mechanical strength of Ta is slightly low (Cossairt 2014). Pure lead (Pb) has a major drawback due to its poor structural characteristics, low melting point of 327 °C, and its toxicity, but reported to be extremely effective as an additive to other materials for improving the ability to block gamma radiation (Cossairt 2014).

In this paper, a conceptual shielding design of metal hydrides multilayers is investigated widely including comparing with a single layer heavy metal extending the previous researches. Neutron absorbers such B, Hf, Cd, and born carbide (B_4 C) are not included in our multilayer model for simplicity. And the Monte Carlo N-Particle code (MCNP6.1) is used for the simulation analysis.

Table 1. Shielding compositions, density and atomic number of individual materials (chemical formal in parenthesis).

Materials	Density (g cm^{-3})	Atomic Mass
lithium hydride (LiH)	0.775	7.948
magnesium hydride (MgH_2)	1.450	26.32
titanium hydride (TiH2)	3.750	49.88
zirconium (II) hydride (ZrH_2)	5.600	93.24
lead (Pb)	11.34	207.20
tantalum (Ta)	16.69	180.94
tungsten (W)	19.30	183.84

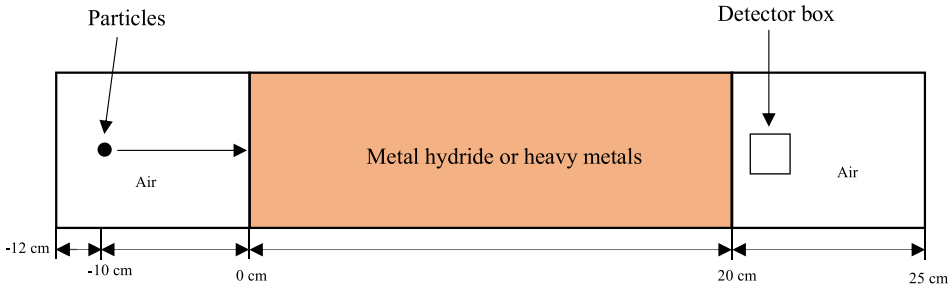


Figure 1. Single-layer model.

The multilayer models of metal hydride and heavy metals by stacking together to shield both neutron and gamma radiation are investigated, and the optimal combination is provided from the results. The simulation code, modeling, and analysis conditions are explained in section 2. The result of single layer and the multilayer analyses are discussed in section 3. Results include the neutron and gamma fluxes under various conditions such as material thickness, particle energy and neutron and gamma spectra. And analysis of displacement per atom (DPA) of individual material is also carried out. Finally, section 4 summarises the findings and describes future work.

2. Modeling and simulation conditions

2.1. Computer code

The MCNP (Monte Carlo N-Particles) code with the ENDF/B-VII.1 neutron cross-section library and MCPLIB84 gamma library is employed for the shielding analysis (Pelowitz 2013). The MCNP code can perform analysis for lots of applications including radiation protection, dosimetry, radiation shielding, nuclear criticality calculations, particle acceleration and charged particle transport, etc. In this study, the simulation of radiation shielding problems uses a general source definition 'SDEF' card combined with F4 tally to obtain the cell flux in the detector region as shown in figure 1. The simulation of entire analysis uses 50 000 000 particle histories. The DPA analysis has been carried out with MCNP, too.

2.2. Model description

The single-layer and multilayer models are shown in figure 1 and figure 2, respectively. The model of multilayer accounts for 2-layers up to 8-layers. Both models are divided into three regions. The width of the first region of the geometric model is 12 cm. Neutron and photon particles propagate in a straight line along the x-axis before entering the target material at -10 cm. It is assumed that the first region is filled with air. The second area is the location of the target material. The target material width is 20 cm by considering mean free path of neutron and gamma rays. The shielding thickness of the target material and the change in particles spectrum are the key parameter to evaluate the effectiveness of shielding material for single—layer model. The small box of the third part of the model is the detector, which artificially contains only air. It detects neutron or gamma ray passing through the shielding materials. It has been found that gamma energy in the reactor is in the range between 0.10 MeV and 10 MeV and neutron energy ranges from 0.18 MeV to 12 MeV (Singh *et al* 2014). To simplify the simulation, the single-layer model with mono-energy range of 1 MeV to 10 MeV is considered in this analysis.

2.3. Multilayer model

The advanced shielding material to reduce the intensity of neutrons and gamma radiation is achieved by mixed hydrogen-rich materials, heavy metal elements, and some other neutron absorber materials (Hu *et al* 2008, Fantidis 2015). The metal hydrides are widely used for hydrogen storage these days (Grochala and Edwards 2004, Reshak 2013, Wang *et al* 2016b, Barthelemy *et al* 2017). Therefore, this article focuses on the conceptual radiation shielding design using the metal hydride multilayers by coupling metal hydride materials and heavy metals in different layers with various geometries. The source energy of 10 MeV is used for multilayer model analysis for both neutron and gamma ray. The main arrangement of multilayer shielding media is that the metal hydride material is placed on the front layer and the heavy metal is placed on the back one as a couple or a pair, i.e. the two layer model in figure 2(a). The hydrogen in the metal hydride compound acts as a moderator which slows down highly energetic neutrons. The interaction of neutrons with metal hydrides could lead to the production of γ from the capture or inelastic scattering reactions, and the heavy metal layer behind can reduce the γ intensity significantly with an increased gamma reaction due to the high atomic number (Z). The couple layers of multilayer model are kept increasing from two layers to four, four to six and six to eight layers while maintaining the positions of metal hydride on the front and heavy metal on the back. In order to study sensitivity analysis of multilayer shielding, the design cases of multilayer geometric models are given as follows and they are depicted in figure 2(a)–(d).

(a) 1st case: two—layer model is composed of 10 cm equally of metal hydride in the 1st layer, and heavy metal in the 2nd layer.
(b) 2nd case: four—layer model is consisted of 5 cm equally of metal hydride in the 1st and the 3rd layer, which 2nd and 4th layer are heavy metal.
(c) 3rd case: six—layer model is designed as 3.33 cm equally. The 1st, 3rd and 5th layer locate metal hydride material, and 2nd, 4th and 6th layer place heavy metal.
(d) 4th case: eight—layer is designed with 2.5 cm equally. The 1st, 3rd, 5th and 7th layer locate metal hydride compound. The 2nd, 4th, 6th and 8th layer contain heavy metal.

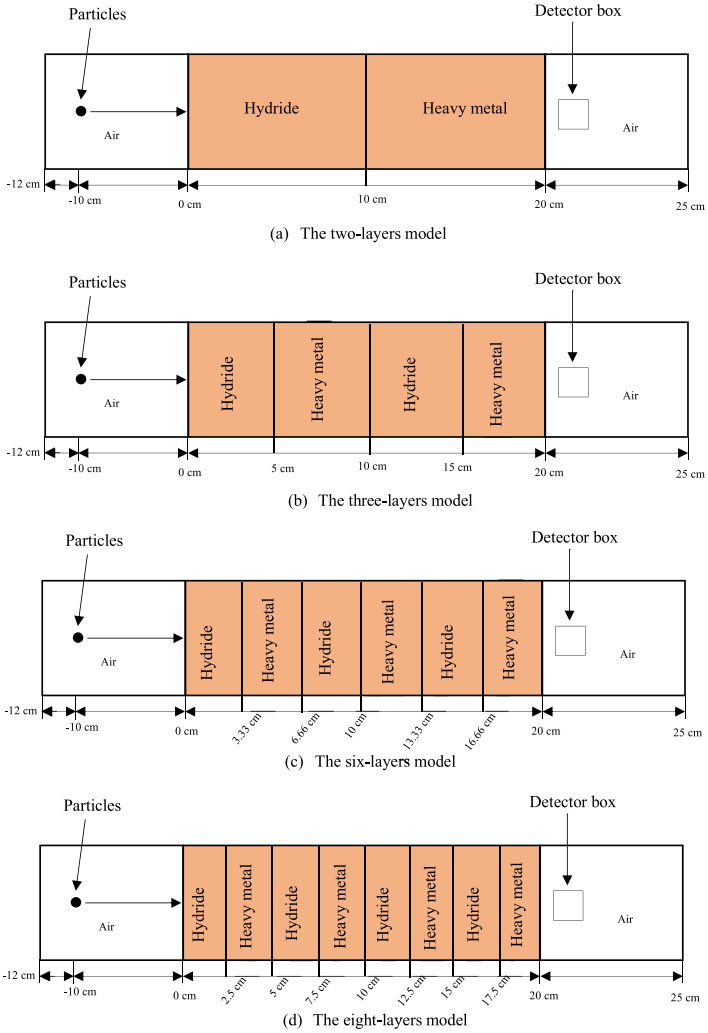


Figure 2. The multilayer model.

(e) 5th case: the unequal thickness of multilayer model, which analysed for TiH_2 and ZrH_2 coupling with W of two—layer model. The denotation is made, i.e. TiH_2_W0218 where '02' means 2 cm of 1st layer and '18' refers to 18 cm of 2nd layer.

The four case studies distribute to be equal material thickness, while the 5th case study allows the variation of unequal thickness of 2—layer model while maintains the total thickness of 20 cm. The opposite positioning by replacing the heavy metal on the front layer and the metal hydride on the back one is also discussed to compare the effectiveness of their shielding performance with those of the references.

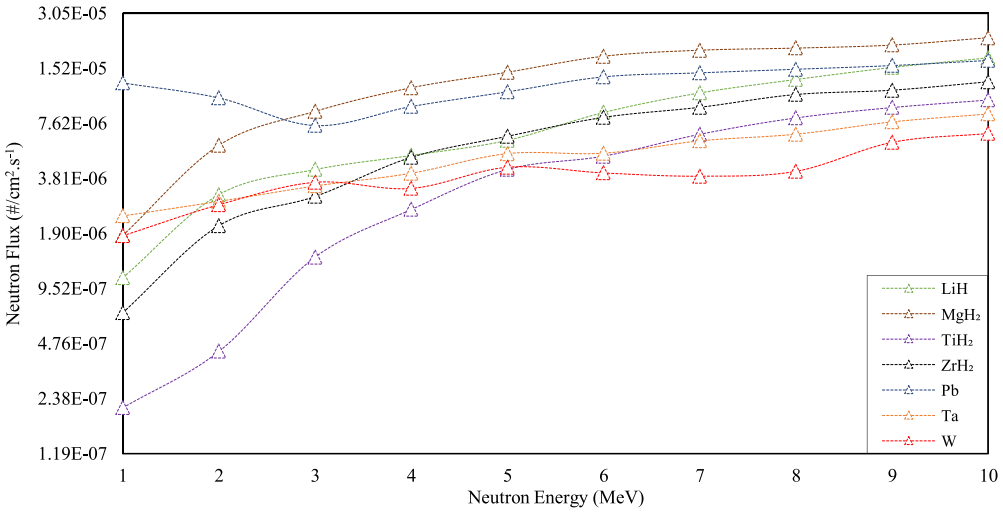


Figure 3. The variation of neutron energy vs. neutron flux of each materials of single-layer model.

3. Results and discussion

3.1. Single-layer analysis

The sensitivity analysis of the single-layer model is carried out in order to investigate the performances of the shielding material candidates. Source energy and shield thickness become the main simulation parameters. Figure 3 and figure 4 show the target material with a fixed 20 cm thickness for the neutron and gamma fluxes relative to the particle energy of 1 MeV to 10 MeV, respectively. In figure 3, the individual materials show smoothly and slightly increase in similar trend as the neutron energy increases. Metal hydrides exhibit less effective shielding capabilities than heavy metals due to their low density of metal hydrides. TiH_2 shows better performance than other materials in the lower source energy range from 1 MeV to 5 MeV, whereas W has a little advantage over other materials at the higher neutron energy range between 5 MeV and 10 MeV. As shown in figure 4, it is clear that heavy metals are more prominent than hydride materials, especially W due to its high density. Unlike Pb, W shows the best and superior performance in suppressing gamma-ray intensity following by Ta. The density of W is higher than other metals (Ta, 13%, Pb, 41%) where the higher density enables the element to be dense and the interaction of scattering particles with the atoms of W is likely to occur more frequently than other elements. And, the curve shows almost constant at the range of the change of gamma energy. ZrH_2 exhibits better shielding of gamma-ray radiation than other hydride materials.

Figure 5 and figure 6 show the neutron and gamma fluxes of target material with different thicknesses at single source energy of 10 MeV, respectively. Obviously, the increase in target thickness indicates a proportional decrease in the flux of emitted particles. As shown in figure 5, at a thickness of 2 cm, all materials have similar neutron flux. However, as the thickness of the target material increases, the neutron flux begins to decrease at a constant rate. In this case, heavy metals such as W and Ta should superior performance to hydride metals, and ZrH_2 shows better effectiveness than other hydrides. Under the same conditions, the gamma flux versus the thickness of target material exhibits comparable to the neutron flux as demonstrated

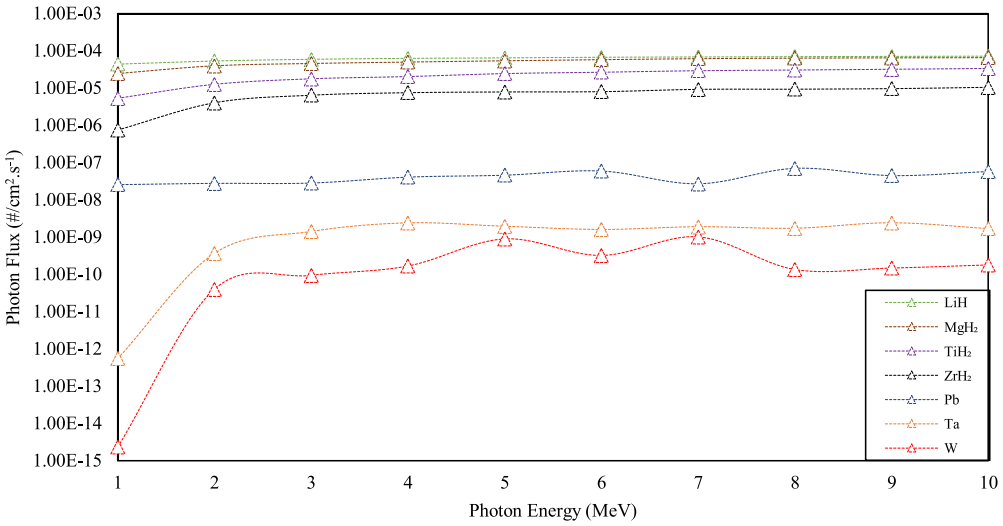


Figure 4. The variation of gamma energy vs. gamma flux of each materials of single-layer model.

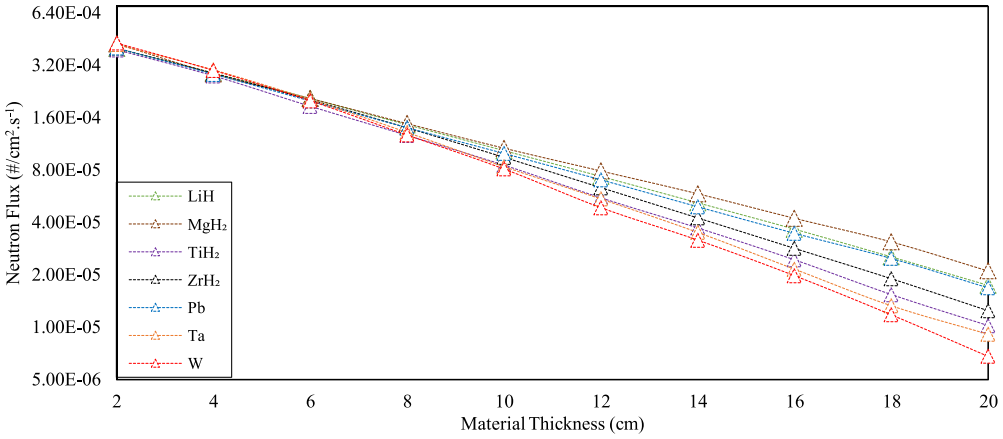


Figure 5. The thickness of each materials vs. neutron fluxes at energy of 10 MeV of single-layer model.

in figure 6. Among the heavy metals, W shows highest strength against the gamma radiation than other materials following by Ta. Moreover, despite the density of ZrH_2 is half lower than Pb, ZrH_2 gives better competence than Pb. The detail information of neutron and gamma flux corresponding to particle energy change and material thickness change is given in appendix.

To summarise results of the single-layer shielding analysis, heavy metals such as W and Ta clearly show better performance in shielding neutrons and gamma-rays. When changing energy of the particles and the thickness of the target material as the same simulation conditions, TiH_2 and ZrH_2 are interchangeable and work superior to LiH and MgH_2. Table 2 list the detailed neutron flux and gamma flux values of the signal-layer model at 1 and 10 MeV, respectively.

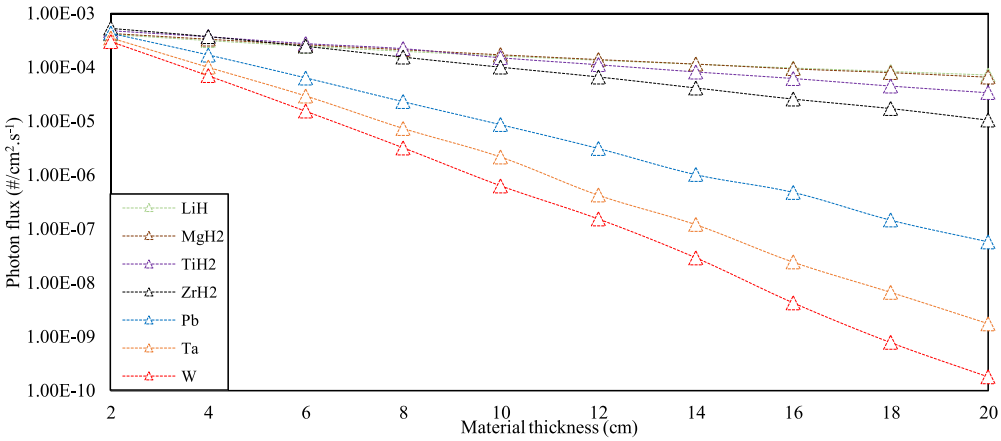


Figure 6. The thickness of each materials vs. gamma fluxes at energy of 10 MeV of single-layer model.

Table 2. Neutron and photo fluxes of single-layer model for various materials (20 cm thickness).

	Neutron flux (#/cm^2.s^{-1})				Photon flux (#/cm^2.s^{-1})			
Materials	1 MeV	Re	10 MeV	Re	1 MeV	Re	10 MeV	Re
LiH	1.09E-06	0.0710	1.74E-05	0.0182	4.41E-05	0.0117	7.22E-05	0.0092
MgH_2	1.86E-06	0.0354	2.11E-05	0.0110	2.48E-05	0.0106	6.58E-05	0.0069
TiH_2	2.12E-07	0.1188	1.02E-05	0.0164	5.36E-06	0.0245	3.38E-05	0.0096
ZrH_2	7.03E-07	0.0597	1.24E-05	0.0147	7.53E-07	0.0617	1.05E-05	0.0174
Pb	1.27E-05	0.0216	1.68E-05	0.0175	2.57E-08	0.7247	5.78E-08	0.3153
Ta	2.38E-06	0.0342	9.12E-06	0.0175	5.49E-13	0.6958	1.70E-09	0.2468
W	1.85E-06	0.0368	6.80E-06	0.0199	2.32E-15	0.7112	1.80E-10	0.2204

Re:Relative error

3.2. Multilayer model analysis

Based on the results of the single-layer model, two metal hydrides, TiH_2 and ZrH_2, and two heavy metals, Ta and W, are chosen for the multilayer analysis following the conceptual ideas of section 2.3. Figure 7 shows the neutron flux of the multilayer model, and table 3 shows the detailed values of these neutron fluxes corresponding to geometrical multilayer model. The neutron flux is obtained in the detector box with F4 tally in MCNP code. From the results, when a couple of multilayers is added, the shielding capabilities of the model is improved significantly, especially W-enhanced TiH_2 for all cases. Figure 8 and table 4 show the gamma flux and its detailed values corresponding to its models, respectively. The multilayer model of the four cases shows that as the number of cases increases, the gamma flux decreases significantly. However, for ZrH_2 adjacent to Ta layer, a different situation is shown displaying the gamma flux jump in the second case higher than in first case. This issue may occur because gamma interaction with materials more in the second case than in the first case. From the first case through the fourth case of ZrH_2 adjacent to W layer, the gamma flux keeps decreasing

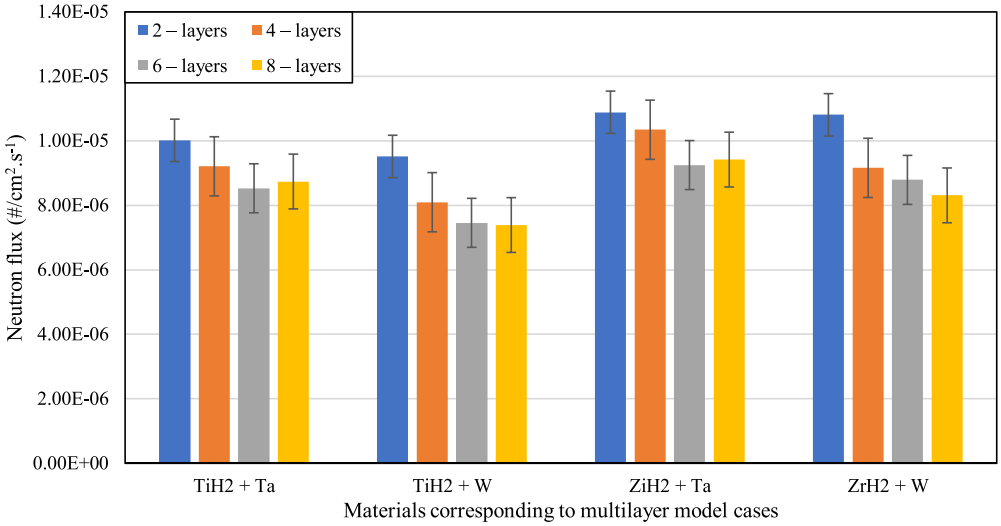


Figure 7. Neutron flux of multilayer model of the main scenario of multilayer models (represent with error bar).

Table 3. Neutron fluxes of multilayer models at constant particle energy of 10 MeV of main scenario.

	Neutron Flux (#/cm^2.s^{-1})							
Material	2—layers	Re	4—layers	Re	6—layers	Re	8—layers	Re
TiH_2 + Ta	8.73E-06	0.0252	8.62E-06	0.0252	8.53E-06	0.0253	8.74E-06	0.0252
TiH_2 + W	7.79E-06	0.0266	7.36E-06	0.0272	7.46E-06	0.0272	7.39E-06	0.0273
ZiH_2 + Ta	1.03E-05	0.0232	1.02E-05	0.0230	9.25E-06	0.0244	9.42E-06	0.0244
ZrH_2 + W	9.24E-06	0.0245	8.94E-06	0.0246	8.79E-06	0.0252	8.31E-06	0.0258

Re: Relative error

as a paired of multilayer material keep increasing from two to eight layers. The coupling multilayer model of ZrH_2 and W gives the best candidate for shielding gamma rays, while TiH_2 and W pairings are good for neutron shielding.

The analysis of unequally thickness of multilayer model is performed on 2—layer model of TiH_2 and ZrH_2 coupling W, as mentioned above. There are 18 case studies to investigate in 2—layer model for both TiH_2 and ZrH_2. Neutron and gamma fluxes of unequal thickness of 2—layer model exhibits in figure 9. The results suggest that, when a thin layer of metal hydride placed in the front of a thick heavy metal, the neutron and gamma shielding performance is improved, contrary when a thick layer of metal hydride placed in the front. By comparing analysis results of single—layer model, the 1st case to 5th case, the case studies of TiH_2_W0218 and ZrH_2_W0218 of 5th case show better performance in term of neutron and gamma-rays shielding.

The results of the opposite scenario of replacing heavy metals on the front layer and metal hydrides on the back layer as a pair multilayer model are presented in figure 10 and figure 11, for neutron and gamma, respectively. The results in figure 10 suggest that, contrary to the main situation of the neutron shielding multilayer model, in the opposite case, the increase of the coupled shielding layers show a slight increase in neutron flux. This probably happens

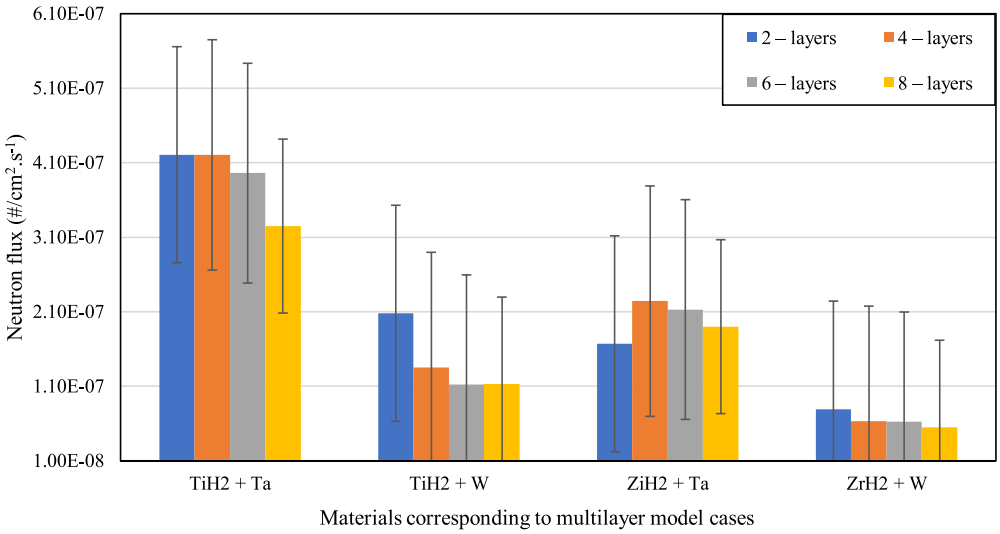


Figure 8. Gamma flux of multilayer model of the main scenario of multilayer models (represent with error bar).

Table 4. Photon fluxes of multilayer models at constant particle energy of 10 MeV of main scenario.

	Photon Flux (#/cm^2.s^{-1})							
Material	2—layers	Re	4—layers	Re	6—layers	Re	8—layers	Re
TiH_2 + Ta	4.21E-07	0.1180	4.21E-07	0.1347	3.96E-07	0.1222	3.25E-07	0.1350
TiH_2 + W	2.08E-07	0.1896	1.35E-07	0.2171	1.12E-07	0.2264	1.13E-07	0.2217
ZiH_2 + Ta	1.67E-07	0.1864	2.24E-07	0.2008	2.13E-07	0.1712	1.90E-07	0.1749
ZrH_2 + W	7.94E-08	0.3304	6.30E-08	0.2964	7.32E-08	0.2922	5.51E-08	0.3099

Re: Relative error

because the front layer is a heavy metal that does not degrade neutron energy well, and when a coupling shielding layers is added, the thickness of shield material become thinner, which results in an increase in neutron throughput pass through. In figure 11, the results of the gamma shielded multilayer model show that only Ta with TiH_2 behaves similarly to neutrons, but the remaining coupling multilayer models exhibit differently. Even though the opposite scheme appears to be contrary to the main scheme, the neutron and gamma fluxes of opposite scheme show slightly lower than that to the main scheme.

To sum up, the 4th case which contains eight—layer model with equal thickness of 2.5 cm, works effectively to shield both neutron and gamma. TiH_2 couple W is good for neutron shielding, and ZrH_2 couple W is good for gamma shielding, respectively. The uneven thickness of multilayer model is also perform on 2—layer model. The thin thickness of metal hydride couple with thick thickness of heavy metal could enhanced the shield capability.

3.3. Displacement per atom (DPA) analysis

Displacement per atom (DPA) is discussed in this section. DPA is a standard parameter to determine the radiation damage in materials, an integral magnitude that includes information

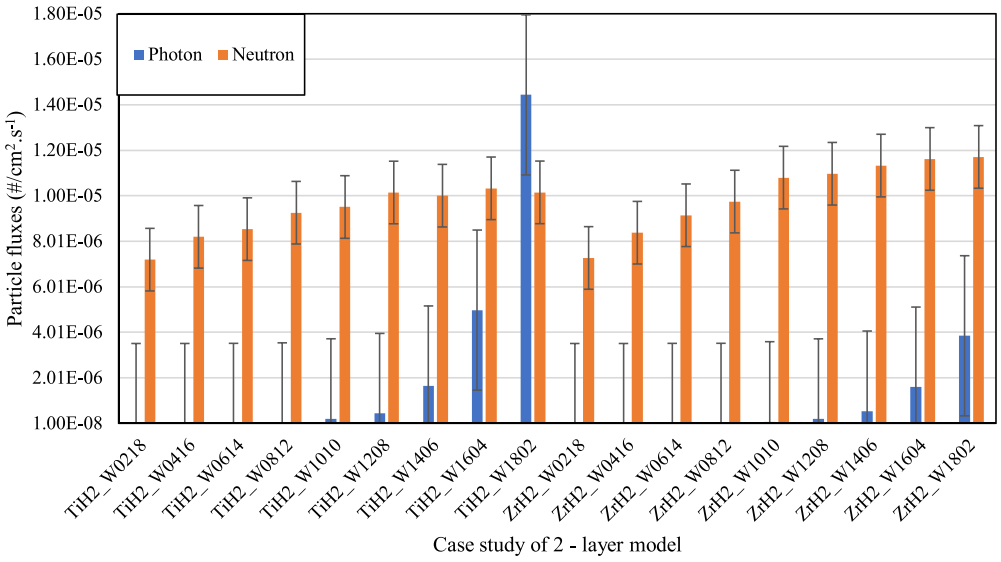


Figure 9. Neutron and gamma flux of unequal thickness of 2—layer model (represent with error bar).

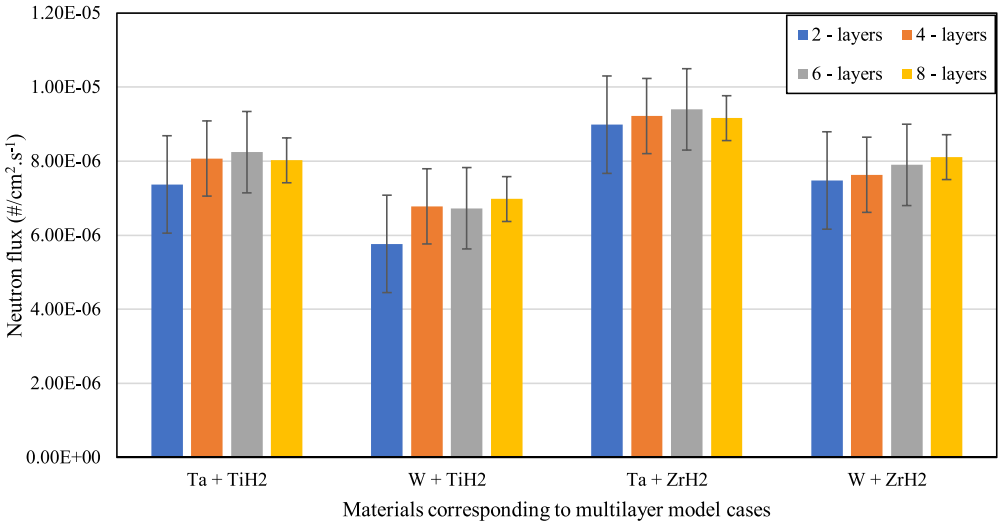


Figure 10. Neutron flux of multilayer model of the opposite scenario of multilayer models (represent with error).

about the material response and the neutron fluence to which the material was exposed (Mascitti and Madariaga 2011). Radiation damage by nuclear collisions and reactions that produced high-energy recoil atoms of host materials or reaction products. These recoil atoms generate electron excitation in the host material, which causes other host atoms to shift, which is call displacement damage. In metals, this can cause permanent damage in the process. Radiation

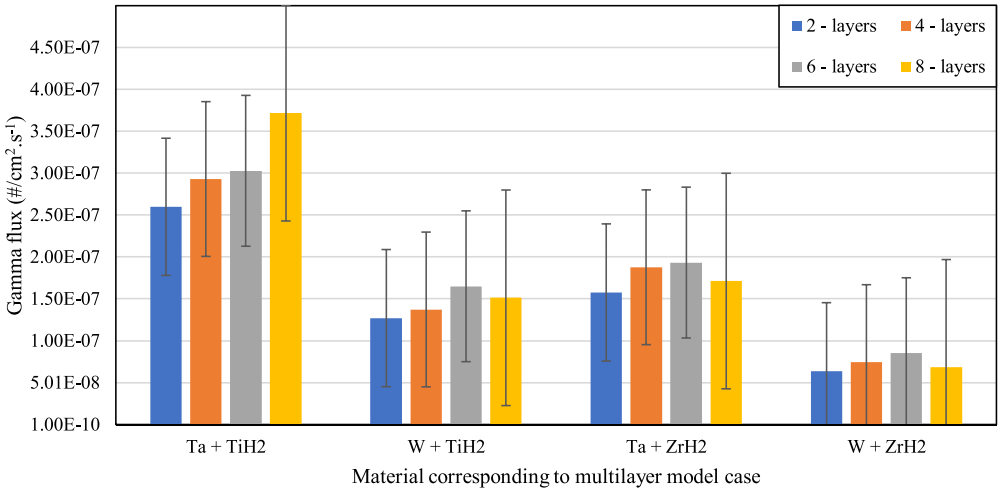


Figure 11. Gamma flux of multilayer model of the main scenario of multilayer models (represent with error bar).

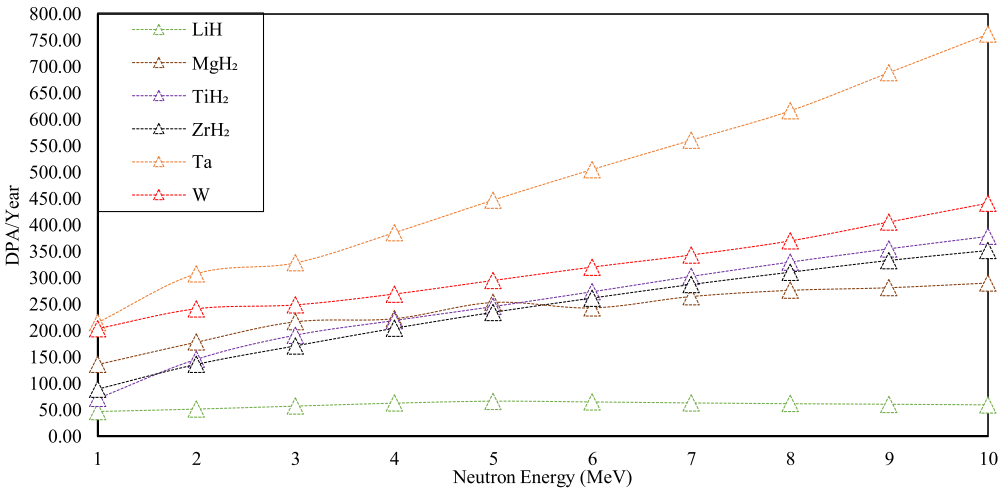


Figure 12. DPA per year of individual materials at 1 and 10 MeV of particle energy of single-layer model.

damage may alter the mechanical properties of materials. The properties that can be effect life-times include ductility, tensile strength, embrittlement, cracks, swelling, elongation irradiation creep, phase transformation, thermal conductivity, thermal expansion (Wootan 2014).

The analysis of DPA per year is calculated only for the neutron particle. The rate of DPA per year is calculated by multiplying the MCNP's FM4 card with DPA neutron cross-section of each material. Figure 12 shows the DPA/year of all single materials with a constant thickness of 20 cm and neutron energy range from 1 MeV to 10 MeV. As the neutron energy increases, LiH exhibits the lowest and constant DPA/year. LiH would be an promising shielding material, such as a space reactor, but the analysis of the signal-layer model clearly shows otherwise. Other materials such as TiH_2, ZrH_2, W and MgH_2 share comparable behavior. However, with the

Table 5. Neutron DPA/Year of single-layer model.

	Materials						
Energy (MeV)	LiH	MgH_2	TiH_2	ZrH_2	Pb	Ta	W
1	47.14	136.11	71.72	89.59	208.93	215.55	204.14
2	51.82	178.49	146.05	136.49	378.85	308.13	241.77
3	57.41	217.00	191.77	171.50	522.21	328.68	249.16
4	63.02	222.98	219.52	204.79	571.78	385.91	269.75
5	66.75	253.69	246.00	235.08	570.88	447.62	295.41
6	65.32	243.44	274.01	262.08	557.79	505.50	320.74
7	63.47	264.88	303.05	287.70	558.22	560.93	343.64
8	62.00	276.62	330.17	311.01	574.06	616.67	370.32
9	60.94	281.64	355.33	333.28	570.76	689.08	406.09
10	59.77	290.52	378.88	352.13	485.50	761.77	441.77

Relative error: between 0.0007 to 0.0008

increasing of neutron energy, atoms of Ta suffer damage from the radiation higher than other materials like W but with lower DPA/year than Ta. The DPA/year of Pb is higher than other material at energy range from 1 MeV to 4 MeV and become stable from 4 MeV to 9 MeV and then drop lower than Ta at 10 MeV. And, table 5 lists the detailed values of DPA/year of individual materials for neutron energy range from 1 MeV to 10 MeV.

4. Conclusion

The multilayer shielding sensitivity with metal hydrides is proposed by using heavy metal reinforcement to enhance the simultaneously shield capability for both neutron and gamma radiation by using MCNP6.1 code. The arrangement of multilayer model is to put the metal hydride in front layer and the heavy metal in the back layer as a pair of multilayer models. A pair of multilayer models is increase up to four times from 2—layers to 8—layers, whereas the thickness of target material is maintained. First study the single-layer model analysis to select suitable and effective shielding material candidates in order to evaluate the multilayer model. According to the simulation results of the single-layer model, two metal hydride TiH_2 and ZrH_2 and the reinforcement materials Ta and W were used in the analysis of multilayer models. The results of multilayer models are summarised as follows.

(a) ZrH_2 and W pairings are the best candidate for shielding gamma at higher energy ranges.
(b) TiH_2 and W pairings are the best candidate for shielding neutron at higher energy ranges.
(c) Comparing the single—layer model of TiH_2 and ZrH_2 to multilayer model of TiH_2 and ZrH_2 adjacent to W layer, shows a significant improvement in shielding capability of TiH_2 and ZrH_2 couple with W to shield neutron and gamma, respectively.
(d) The thin layer of metal hydride pair a thick layer of heavy metal could enhanced the shielding performance.
(e) The DPA analysis shows that LiH would be an ideal shieling material with very low DPA/year and constantly stable at lower to higher particle energy. And, the DPA/year of TiH_2 and ZrH_2 shows an approximately 6 time higher than that to LiH. Both TiH_2 and ZrH_2 could be used instead of LiH if needed.

Table A1. Neutron fluxes of single-layer model for various material (20 cm material thickness).

Neutron flux of various materials corresponding to energy change (#/cm^2.s^{-1})							
Neutron energy (MeV)	LiH	MgH_2	TiH_2	ZrH_2	Pb	Ta	W
1	1.09E-06	1.86E-06	2.12E-07	7.03E-07	1.27E-05	2.38E-06	1.85E-06
2	3.11E-06	5.79E-06	4.31E-07	2.10E-06	1.05E-05	2.86E-06	2.74E-06
3	4.26E-06	8.87E-06	1.42E-06	3.04E-06	7.42E-06	3.46E-06	3.63E-06
4	5.09E-06	1.19E-05	2.58E-06	4.97E-06	9.43E-06	4.07E-06	3.36E-06
5	6.16E-06	1.45E-05	4.27E-06	6.48E-06	1.13E-05	5.21E-06	4.39E-06
6	8.75E-06	1.78E-05	5.05E-06	8.23E-06	1.37E-05	5.24E-06	4.08E-06
7	1.12E-05	1.92E-05	6.64E-06	9.34E-06	1.44E-05	6.12E-06	3.92E-06
8	1.33E-05	1.97E-05	8.16E-06	1.10E-05	1.51E-05	6.66E-06	4.17E-06
9	1.54E-05	2.04E-05	9.30E-06	1.16E-05	1.58E-05	7.77E-06	6.00E-06
10	1.74E-05	2.11E-05	1.02E-05	1.24E-05	1.68E-05	9.12E-06	6.80E-06

Relative error: between 0.02 to 0.01

Table A2. Photon fluxes of single-layer model for various material (20 cm material thickness).

Photon flux of various materials corresponding to energy change (#/cm^2.s^{-1})							
Photon energy (MeV)	LiH	MgH_2	TiH_2	ZrH_2	Pb	Ta	W
1	4.41E-05	2.48E-05	5.36E-06	7.53E-07	2.57E-08	5.49E-13	2.32E-15
2	5.35E-05	3.98E-05	1.27E-05	4.05E-06	2.78E-08	3.68E-10	4.00E-11
3	5.95E-05	4.60E-05	1.78E-05	6.46E-06	2.84E-08	1.43E-09	9.47E-11
4	6.31E-05	5.01E-05	2.05E-05	7.55E-06	4.09E-08	2.42E-09	1.68E-10
5	6.50E-05	5.43E-05	2.46E-05	7.91E-06	4.61E-08	1.97E-09	8.98E-10
6	6.75E-05	5.82E-05	2.68E-05	8.05E-06	5.96E-08	1.61E-09	3.24E-10
7	6.88E-05	6.18E-05	2.94E-05	9.28E-06	2.71E-08	1.91E-09	1.02E-09
8	6.99E-05	6.39E-05	3.07E-05	9.38E-06	7.04E-08	1.74E-09	1.36E-10
9	7.13E-05	6.51E-05	3.21E-05	9.71E-06	4.51E-08	2.44E-09	1.48E-10
10	7.22E-05	6.58E-05	3.38E-05	1.05E-05	5.78E-08	1.70E-09	1.80E-10

Relative error: between 0.08 to 0.56

It is concluded that multilayer model of metal hydride, especially TiH_2 or ZrH_2 coupling with W, would simultaneously increase the shielding ability to shield neutron and gamma rays, even though these materials would suffer radiation damage in its atoms. It is also found out that the thin layer of metal hydride pair with a thick layer of heavy metal could potentially increase the shielding capability for both neutron and gamma spontaneously.

The results in this study are helpful to develop new and innovative shielding materials for future industrial applications including nuclear facilities, fusion reactors, and other radiation applications that require advanced shielding materials. Furthermore, detail analysis including experimental tests are expected to validate performance of multilayer shielding before applying real situations.

Table A3. Neutron fluxes of single-layer model for various material (10 MeV of neutron energy).

Neutron flux of various materials corresponding to thickness of material change (#/cm^2.s^{-1})							
Material Thickness (cm)	LiH	MgH_2	TiH_2	ZrH_2	Pb	Ta	W
2	4.01E-04	4.27E-04	3.94E-04	4.02E-04	4.02E-04	4.21E-04	4.31E-04
4	2.90E-04	2.87E-04	2.79E-04	2.89E-04	2.85E-04	3.02E-04	3.00E-04
6	2.07E-04	2.07E-04	1.86E-04	2.03E-04	2.00E-04	2.04E-04	2.00E-04
8	1.46E-04	1.48E-04	1.27E-04	1.40E-04	1.40E-04	1.32E-04	1.27E-04
10	1.04E-04	1.07E-04	8.55E-05	9.49E-05	9.98E-05	8.38E-05	8.13E-05
12	7.37E-05	7.92E-05	5.55E-05	6.35E-05	7.04E-05	5.47E-05	4.87E-05
14	5.20E-05	5.86E-05	3.73E-05	4.25E-05	4.93E-05	3.51E-05	3.17E-05
16	3.66E-05	4.23E-05	2.44E-05	2.84E-05	3.47E-05	2.16E-05	1.98E-05
18	2.54E-05	3.10E-05	1.54E-05	1.90E-05	2.49E-05	1.33E-05	1.18E-05
20	1.74E-05	2.11E-05	1.02E-05	1.24E-05	1.68E-05	9.12E-06	6.80E-06

Relative error: between 0.008 to 0.04

Table A4. Photon fluxes of single-layer model for various material (10 MeV of photon energy).

Photon flux of various materials corresponding to thickness of material change (#/cm^2.s^{-1})							
Material Thickness (cm)	LiH	MgH_2	TiH_2	ZrH_2	Pb	Ta	W
2	4.10E-04	4.26E-04	4.79E-04	5.29E-04	4.29E-04	3.51E-04	3.00E-04
4	3.14E-04	3.33E-04	3.70E-04	3.72E-04	1.70E-04	1.00E-04	6.96E-05
6	2.48E-04	2.63E-04	2.76E-04	2.46E-04	6.37E-05	2.94E-05	1.53E-05
8	1.99E-04	2.13E-04	2.22E-04	1.55E-04	2.30E-05	7.34E-06	3.21E-06
10	1.63E-04	1.71E-04	1.51E-04	1.01E-04	8.62E-06	2.16E-06	6.33E-07
12	1.36E-04	1.40E-04	1.12E-04	6.64E-05	3.13E-06	4.25E-07	1.54E-07
14	1.15E-04	1.15E-04	8.24E-05	4.15E-05	1.02E-06	1.20E-07	2.93E-08
16	9.74E-05	9.40E-05	6.21E-05	2.58E-05	4.78E-07	2.43E-08	4.26E-09
18	8.39E-05	7.98E-05	4.50E-05	1.72E-05	1.46E-07	6.68E-09	7.80E-10
20	7.22E-05	6.58E-05	3.38E-05	1.05E-05	5.78E-08	1.75E-09	1.80E-10

Relative error: between 0.08 to 0.45

Appendix

The detail information of neutron and photon flux of single layer model for 1) particle energy change (with a fixed material thickness) and 2) material thickness change (with a fixed particle energy) are presented here. Table A1 and table A2 corresponding to figure 3 and figure 4, respectively for neutron and photon flux variation due to the neutron energy change with a fixed 20 cm of material thickness. Table A3 and table A4 corresponding to the figure 5 and figure 6, respectively for neutron and photon flux variation due to the material thickness change with a fixed 10 MeV of particle energy.

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